Tae-Yoon Kim

Undergraduate Program · School of Computing, KAIST

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Interests

Artificial Intelligence, Large Language Models, Cyber-Physical Systems, Defense Science, Security Systems

Education

Korea Advanced Institute of Science and Technology — Expected Feb 2029
B.S. in Computer Science, School of Computing — Daejeon, Korea
GPA 4.11 / 4.3 · Coursework: Machine Learning, Intro. to AI, Computer Organization, System Programming, Data Structures

University of California, Berkeley — Jan 2027 – May 2027
Exchange Student, UCEAP Reciprocity Program — Berkeley, CA

Gwangju Seongdeok High School — Graduated Jan 2025
High School Diploma; study abroad at St. Francis Catholic Secondary School, Ontario, Canada (Jan 2023) — Gwangju, Korea

Research Experience

Machine Learning & Vision Lab, KAIST — Sep 2026 – Present
Individual Study · Advisor: Prof. Hyunwoo J. Kim — Daejeon, Korea

- Developing Gramian SVD Checkpoint Compaction, a low-rank state compression method for prefix caching in linear-attention models, with GPU experiments toward a paper submission.
- Learned modern LLM inference design by studying frontier architectures and FlashAttention, covering KV caching, sparse attention and MoE routing; tested a self-proposed KV cache–sparse indexer design and ruled it out through analysis and small-scale experiments.

Urban Robotics Lab, KAIST — Jun 2026 – Aug 2026
Individual Study · Advisor: Prof. Hyun Myung — Daejeon, Korea

- Learned the ROS development workflow: building nodes in Docker, simulating a TurtleBot3 in Gazebo, and visualizing LiDAR, IMU and RGB-D streams in RViz.
- Gained hands-on experience with LiDAR SLAM (GMapping, Hector) and autonomous navigation, and with classical vs. learned feature matching by benchmarking seven keypoint pipelines from ORB and SIFT to SuperPoint.

Honors & Awards

Dean’s List, School of Freshman — KAIST, Fall 2025

Freshman Oblige Project Dean’s Award — KAIST, Mar 2026

Color Your 2025 Project Dean’s Award — KAIST, Dec 2025

Highest Honor for Academic Excellence — Seongdeok HS, Jan 2025

Meritorious Service Award — Seongdeok HS, Jan 2025

Teaching

Undergraduate Teaching Assistant — Feb 2026 – Present
Introduction to Programming (CS.10001), KAIST — Daejeon, Korea

- Support Python instruction for first-year students through Q&A.

JuniorKAIST Teaching Tutor — Mar 2026 – Present
Global Institute For Talented EDucation (GIFTED), KAIST — Daejeon, Korea

- Lead C programming sessions and review students’ assignments.

Leadership

President, Club Association — Mar 2023 – Feb 2024
Student Council, Gwangju Seongdeok High School

President, Student System Programming Committee — Mar 2023 – Feb 2024
Club Association, Gwangju Seongdeok High School

Skills

Programming	C, Python, Java, JavaScript, PHP, MySQL
Tools	ROS, Gazebo, Docker
Languages	Korean (native), English (TOEIC 880, DET 115)